HEMPCRETE IS A SUSTAINABLE MATERIAL

SEMINAR REPORT

Submitted by

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CERTIFICATE

This is to certify that the seminar report entitled **'HEMPCRETE IS A SUSTAINABLE MATERIAL'** submitted by **'ARYAN O S (SNM23CE003)'** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Civil Engineering is a bonafide record of the seminar work carried out by her under our guidance and supervision. This report in any form has not been submitted to any other universities or institutions for any purpose.

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ACKNOWLEDGEMENT

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ABSTRACT

Hempcrete, a bio-based composite made primarily from hemp shives and a mineral binder, is one such material being investigated for sustainable building construction. It differs fundamentally from conventional structural concrete. It is lightweight and porous, and its principal current engineering role is as a non-load-bearing infill, insulation or envelope material. The porous structure contributes to low thermal conductivity, acoustic absorption and moisture buffering. Hemp cultivation also offers a pathway for biogenic carbon storage because the growing plant removes carbon dioxide from the atmosphere and incorporates carbon into its biomass.

The reviewed evidence also demonstrates important limitations. Mechanical strength is comparatively low, material properties vary with hemp-shive characteristics and binder formulation, and standardized design procedures are still developing. Moisture management, curing, material compatibility and long-term durability must be considered carefully. Fire performance is promising in reported studies but remains dependent on formulation and test configuration. The environmental advantage of hempcrete also depends on the complete life cycle, including cultivation, processing, binder manufacture, transportation, construction and end-of-life.

CONTENTS

TITLE	PAGE NO
Acknowledgement	I
Abstract	II
Contents	III
List of Figures	V
CHAPTER 1 INTRODUCTION	1
1.1 Need for Sustainable Building Materials	2
1.2 Hempcrete as a Building Material	2
1.3 Objectives	2
1.4 Scope	2
CHAPTER 2 LITERATURE SURVEY	3
CHAPTER 3 MATERIALS, PROPERTIES AND METHODOLOGY	5
3.1 Hemp Anatomy and Raw Material	5
3.2 Hemp Shives	5
3.3 Significance of Hemp Stem Anatomy	6
3.4 Binder Systems	7
3.5 Mix Design	7
3.6 Construction Techniques	7
3.7 Fresh-State Properties	9
3.8 Mechanical Properties	9
3.9 Thermal Properties	9
3.10 Acoustic Properties	10
3.11 Moisture and Durability	10
3.12 Fire Behaviour	10
3.13 Construction Quality and Fire Safety	10
3.14 Material Compatibility and Interfaces	11
3.15 Limitations of Current Test Data	11

CHAPTER 4 ADVANTAGES AND DISADVANTAGES 12
4.1 Advantages of Hempcrete 12
4.2 Disadvantages of Hempcrete 12
CHAPTER 5 RESULTS AND DISCUSSION 14
5.1 Compatibility with Timber 14
5.2 Compatibility with Masonry and Steel 14
5.3 Real-World Application: Hemp Hotel 15
5.4 Adnams Brewers Distribution Centre 16
5.5 Pierre Chevet Sports Hall 17
5.6 Hygrothermal Performance in Different Climates 18
5.7 Environmental Performance 20
5.8 Life-Cycle Assessment 20
5.9 Economic Considerations 20
5.10 Science Museum Group Storage Facility 20
5.11 Interpretation of Real-World Applications 21
CHAPTER 6 CONCLUSION 22
6.1 Conclusion 22
6.2 Future Scope 22
REFERENCES 23

LIST OF FIGURES

SL NO	TITLE	PAGE NO
3.1	Industrial hemp	5
3.2	Hemp shives used as the plant-based aggregate	6
3.3	Spray-applied hempcrete process	8
3.4	In-situ hempcrete during wall construction	8
3.5	Prefabricated hempcrete blocks (a) and block-wall construction (b)	9
5.1	Hempcrete blocks utilised in a timber post-and-beam structure	14
5.2	Hempcrete cast against masonry (a) and used with a metal structural system (b)	15
5.3	Hemp Hotel under construction, Cape Town	15
5.4	Interior construction of the Hemp Hotel	16
5.5	Adnams Brewers Distribution Centre: exterior (a) and in-situ casting (b)	17
5.6	Pierre Chevet Sports Hall, France	18
5.7	Climate zones represented in the reviewed performance studies	19
5.8	Annual overheating hours reported for the compared buildings	19

CHAPTER 1
INTRODUCTION

Hempcrete is a bio-based construction material produced by combining the woody inner core of the hemp plant, known as hemp hurd or hemp shiv, with a lime-based binder and water. Unlike conventional concrete, hempcrete is not primarily used as a structural load-bearing material. It is mainly used as an insulating infill material for walls, roofs, floors, and other building components. It is considered a sustainable material because hemp is a rapidly renewable crop that requires relatively few resources for cultivation. During its growth, hemp absorbs carbon dioxide from the atmosphere through photosynthesis. When hemp-based materials are incorporated into buildings, a portion of this carbon can remain stored within the building for an extended period. Hempcrete also provides good thermal and acoustic insulation, moisture regulation, and fire resistance, which can contribute to healthier and more energy-efficient buildings.

The construction industry is one of the largest consumers of energy and raw materials, and conventional materials such as cement and steel carry a high embodied carbon cost. This has increased interest in renewable, low-carbon alternatives that can reduce the environmental footprint of buildings without compromising comfort or safety. Hemp grows quickly, can be integrated into crop rotation, and yields a woody core that would otherwise be a by-product of fibre production. When this core is combined with a mineral binder, the resulting composite is light, porous and vapour-permeable, and it can be cast in place, prefabricated into blocks and panels, or sprayed onto walls and ceilings.

Research on hempcrete has grown steadily in recent years, covering binder chemistry, mix design, mechanical and thermal behaviour, moisture performance, fire safety, life-cycle assessment and real-world applications. Projects in Europe, the United Kingdom and South Africa show that hempcrete can be used not only in small houses but also in commercial and multi-storey buildings when it is combined with a suitable structural frame. At the same time, published results vary widely because mixes, materials and test methods differ, and standardized design procedures are still developing. This seminar report therefore reviews the literature on hempcrete as a sustainable building material, covering its materials and binder systems, mix design, construction, properties, case studies and limitations.

1.1 NEED FOR SUSTAINABLE BUILDING MATERIALS

A building material should ideally provide adequate performance while reducing unnecessary environmental impacts. The environmental performance of a building is influenced by operational energy and by embodied impacts associated with raw-material extraction, manufacturing, transport, construction, maintenance and disposal. Bio-based materials are receiving attention because renewable biomass can be incorporated into long-lived construction products while potentially storing atmospheric carbon. Materials that reduce heating and cooling demand, regulate indoor humidity and can be reused or returned safely to the environment at the end of their life are therefore of growing interest to designers and civil engineers.

1.2 HEMPCRETE AS A BUILDING MATERIAL

Hempcrete is a bio-based composite generally made using hemp shives, a mineral binder and water. The hemp shives act as a lightweight porous component, while the binder forms a mineral matrix around the particles. Unlike structural concrete, hempcrete is normally used as non-load-bearing infill, insulation or an envelope component. The principal review emphasizes that its engineering significance arises from the combination of low density, thermal insulation, moisture buffering, acoustic behaviour and potential carbon storage. It can be cast in place around a timber or steel frame, prefabricated as blocks and panels, or sprayed onto walls, and it is usually finished with lime plasters or renders that allow the wall to breathe.

1.3 OBJECTIVES

The objectives of this seminar are to study the composition, binder systems, mix design and construction techniques of hempcrete; evaluate its mechanical, thermal and acoustic properties, moisture behaviour, durability and fire resistance; examine life-cycle implications and real-world projects; and identify its advantages, limitations and research needs. The seminar also aims to show how hempcrete should be specified as part of a complete wall system.

1.4 SCOPE

The report focuses on hempcrete used in buildings, especially walls, infill, insulation, ceilings and envelope systems. Structural use is discussed primarily as a future research possibility because the current literature generally positions hempcrete as a non-load-bearing material. The analysis follows the supplied 2025 review and the scientific studies discussed within it.

CHAPTER 2
LITERATURE SURVEY

A Comprehensive Review of Hempcrete as a Sustainable Building Material – Steyn et al.

Steyn et al. (2025) presented a comprehensive review of hempcrete as a sustainable building material. The study covers its materials, binders, mix design, construction techniques, mechanical and thermal properties, acoustic performance, durability, environmental benefits and applications. The review highlights that hempcrete is mainly suitable as a lightweight, non-load-bearing building-envelope and insulation material. It also identifies carbon sequestration, thermal insulation and moisture regulation as important sustainability benefits. However, standardization, long-term durability, mechanical strength and cost remain important challenges for wider adoption.

Lime Hempcrete Composites Based on Unfired Binders – Haik et al.

Haik et al. (2025) reviewed lime-hempcrete composites produced using unfired mineral binders. The study focuses on the environmental and technical performance of different binder systems used with hemp. It indicates that suitable alternative binders can improve the sustainability of hempcrete by reducing energy consumption and carbon emissions associated with binder production. The research also considers thermal behaviour, mechanical performance and life-cycle aspects. The study demonstrates the importance of selecting an appropriate binder to achieve a balance between strength, insulation and environmental performance.

Low-Carbon Calcined Clay-Based Binders for Sustainable Hempcrete – Nair et al.

Nair et al. (2025) investigated low-carbon binder systems for hempcrete using calcined clay, limestone and ground-granulated blast-furnace slag (GGBFS). Four binder systems were studied and their compressive strength, density, sound absorption, thermal conductivity and surface bond strength were evaluated. The geopolymer hempcrete achieved approximately four times the compressive strength of the reference hempcrete, while the limestone–

calcined-clay system showed about 60% higher compressive strength than the reference mixture. The study demonstrates that alternative low-carbon binders can significantly improve the performance and sustainability of hempcrete.

A Review of the Properties of Hempcrete for Green Building Applications – Jami et al.

Jami et al. (2025) conducted a comprehensive review of the properties of hemp concrete for green building applications. The study reviewed research related to hemp properties, binder characteristics, mechanical behaviour, durability, moisture-related properties, thermal performance, environmental performance, manufacturing processes and building applications.

Mechanical and Thermal Properties of Lime and Hemp Concrete Manufactured by a Projection Process – Elfordy et al.

Elfordy et al. (2024) investigated the mechanical and thermal properties of lime and hemp concrete manufactured using a projection process. The researchers produced hempcrete blocks using lime and hemp shives and examined properties such as density, thermal conductivity, flexural strength, compressive strength and hardness.

SUMMARY OF LITERATURE SURVEY

The five journal studies demonstrate the development of hempcrete research from basic material characterization and manufacturing studies toward advanced low-carbon binder systems and whole-building sustainability assessment.

CHAPTER 3
MATERIALS, PROPERTIES AND METHODOLOGY

3.1 HEMP ANATOMY AND RAW MATERIAL

Industrial hemp is scientifically classified as *Cannabis sativa*. The principal review describes the crop as a rapidly growing plant that can reach approximately one to five metres within a cropping cycle of about 90–120 days. The hemp stem contains six principal anatomical layers: epidermis, cortex, phloem, cambium, xylem and pith. The inner woody component, known as hurd or shive, is the part most directly associated with hempcrete production.



Fig 3.1 Industrial hemp

The stem structure explains why different portions of hemp have different industrial uses. Bast fibres occur toward the outside of the cambium and are important in fibre applications, while the inner woody hurd is the principal particulate component used in hempcrete. The production process therefore begins with separating the woody core from the other botanical components.

3.2 HEMP SHIVES

Hemp shives are lightweight, porous and hygroscopic. Their particle size and shape influence the void structure, binder distribution, density and thermal behaviour of the final composite. The material is therefore sensitive to raw-material consistency. A change in shive grading can alter packing and the amount of binder required to achieve a continuous matrix.



Fig 3.2 Hemp shives used as the plant-based aggregate

3.3 SIGNIFICANCE OF HEMP STEM ANATOMY

The anatomy of the stem is directly related to the way hemp is processed for construction. The outer bast region contains fibres that are valuable for textile and composite applications, while the inner woody hurd is the component that is separated, processed and reduced into shives for hempcrete. The cambium acts as a transition zone between these regions. Understanding this structure is therefore useful because the quality and characteristics of the hurd influence the final composite.

The principal review also notes that hemp can be harvested within a relatively short cropping cycle of about 90–120 days. After harvesting, the woody core is separated from the other botanical components. The resulting shives are not an inert aggregate in the conventional sense. They are porous, hygroscopic and capable of interacting with both water and the mineral binder. Their internal pore structure contributes to the low density and thermal behaviour of hempcrete, while their moisture response influences curing and long-term hygrothermal performance.

The review also discusses hemp harvesting and processing. Hemp can be incorporated into crop rotation, and the literature reports comparatively modest requirements for fertilization and plant-protection products. After harvesting, the woody core is separated and processed into a construction-grade particulate material.

3.4 BINDER SYSTEMS

Binders form the mineral matrix around the hemp shives. Lime is the most common binder category discussed in the literature. Natural hydraulic lime can develop strength through hydraulic reactions, while hydrated lime gains strength through carbonation. Pozzolanic additions can alter reaction products and performance. Alternative low-carbon and geopolymer-type systems have also been investigated.

Binder selection influences density, curing, mechanical strength, thermal conductivity, acoustic response and environmental impact. This is one reason why published hempcrete properties cannot be treated as universal constants. A reported strength or conductivity value must always be interpreted together with the mix composition, density, age and testing method.

Table 3.1 Main constituents of hempcrete

Constituent	Role	Important variables
Hemp shives	Lightweight porous aggregate	Particle size, grading, moisture
Mineral binder	Matrix and bonding	Binder chemistry, content, curing
Water	Workability and reaction	Initial moisture, quantity, drying
Optional additions	Property modification	Pozzolans, water retainers, other modifiers

3.5 MIX DESIGN

Unlike conventional structural concrete, hempcrete has no single universally accepted mix proportion. The final material depends on the desired density, thermal performance, construction technique, binder characteristics, water content, hemp-shive properties, compaction and curing. Increasing density can improve some mechanical properties but can also increase thermal conductivity. The design problem is therefore a balance between strength, insulation, moisture behaviour and constructability.

3.6 CONSTRUCTION TECHNIQUES

Common construction approaches include in-situ casting, prefabricated blocks, panels and spray application. In-situ casting involves mixing hemp shives and binder at the construc-

tion site and placing the material into temporary or permanent formwork around a structural frame. Prefabricated blocks are manufactured and cured away from the building site. Spray application can create a continuous layer and is particularly relevant to renovation and complex surfaces.

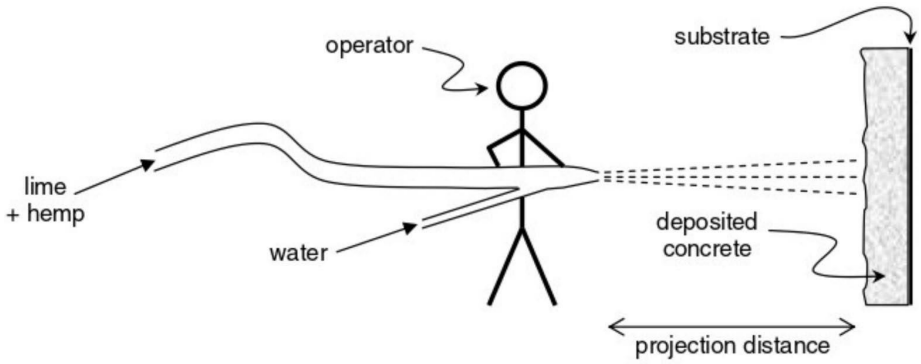


Fig 3.3 Spray-applied hempcrete process



Fig 3.4 In-situ hempcrete during wall construction



(a) (b)

Fig 3.5 Prefabricated hempcrete blocks (a) and block-wall construction (b)

3.7 FRESH-STATE PROPERTIES

Fresh hempcrete is lightweight and porous and is not designed around the high-flow placement and early compressive strength associated with conventional concrete. Workability depends on water content, binder type, shive characteristics and mixing sequence. Appropriate tamping or compaction is required for uniformity, but excessive compaction can reduce porosity and change thermal performance.

3.8 MECHANICAL PROPERTIES

Mechanical strength is one of the principal limitations of hempcrete. The porous lightweight structure produces much lower compressive strength than structural concrete. Consequently, hempcrete is generally used around a structural frame. The review nevertheless reports that binder content, hemp-to-binder ratio, density, hemp treatment and binder chemistry can influence strength. Research is therefore focused on improving strength without losing the low-density advantages.

3.9 THERMAL PROPERTIES

Thermal insulation is one of hempcrete's strongest engineering advantages. Its porous structure and low density reduce heat transfer. The principal review reports that many hempcrete thermal-conductivity values are lower than those of autoclaved aerated concrete at comparable densities. Moisture, density, binder composition and particle characteristics can all influence the measured value.

3.10 ACOUSTIC PROPERTIES

The porous internal structure of hempcrete can absorb sound energy. The principal review indicates that smaller hemp shives and lime-pozzolanic binders can produce higher sound absorption in selected studies, while other investigations identify the quantity of hemp shives as a major determinant. Acoustic performance is therefore dependent on material composition, density, thickness and assembly configuration.

3.11 MOISTURE AND DURABILITY

Hempcrete is hygroscopic and can absorb and release water vapour as surrounding humidity changes. This moisture-buffering behaviour can stabilize indoor relative humidity. However, the same moisture responsiveness requires appropriate wall detailing. Prolonged liquid-water exposure, inadequate drying paths, poor interfaces or unsuitable finishes can create durability problems.

3.12 FIRE BEHAVIOUR

The principal review reports favourable fire-resistant characteristics in several investigated hempcrete systems. The mineral binder limits the flammability of the plant component, while density, binder chemistry, coatings and wall assembly influence the final fire response. Because fire safety is regulatory, project-specific testing and compliance remain necessary.

3.13 CONSTRUCTION QUALITY AND FIRE SAFETY

Fire performance cannot be considered independently of construction quality. The mineral binder, hemp content, density, moisture condition, surface finish and wall assembly all influence how heat and flame interact with the material. A laboratory result for one formulation should therefore not be transferred directly to another wall system. Openings, service penetrations, joints and surface coatings can also change the behaviour of the completed assembly. For practical construction, the engineer should verify that the specified material reaches the intended density and that the hemp shives are uniformly distributed. Excessive voids, local segregation or uncontrolled moisture can produce variations in the finished wall. Drying is particularly important because hempcrete is normally installed with significant moisture and requires time to reach its service condition. Temporary weather protection and a suitable drying path are therefore part of quality control.

3.14 MATERIAL COMPATIBILITY AND INTERFACES

The interface between hempcrete and the structural frame is another important consideration. Timber, masonry, concrete and steel can all be used in building systems containing hempcrete, but the interface must account for vapour movement, thermal bridging and chemical compatibility. The high alkalinity of lime-based binders requires particular attention around vulnerable steel components. Where direct contact cannot be avoided, corrosion protection or physical separation should be considered.

3.15 LIMITATIONS OF CURRENT TEST DATA

Published results vary because researchers use different hemp shives, binders, densities, curing periods and testing methods. This variation makes direct numerical comparison difficult. Future standards should therefore define material conditioning, specimen geometry, density classes and test procedures so that reported values can be compared reliably across studies and projects.

CHAPTER 4
ADVANTAGES AND DISADVANTAGES

4.1 ADVANTAGES OF HEMPCRETE

Excellent thermal insulation – The porous structure of hempcrete provides good thermal insulation and helps reduce heat transfer through walls, thereby lowering heating and cooling requirements.

Low environmental impact – Hemp is a renewable material that requires relatively little water and pesticide for cultivation. Hempcrete can also store significant amounts of CO_2 during its life cycle.

Lightweight material – Hempcrete has a low density, which reduces the load transferred to foundations and makes handling and construction easier.

Moisture regulation – Hempcrete is vapour-permeable and can absorb and release moisture, helping to regulate indoor relative humidity and reduce condensation.

Good acoustic performance – Its porous structure provides sound absorption, making hempcrete useful for improving acoustic comfort in buildings.

Fire resistance – Studies reviewed in the literature indicate that hempcrete can resist ignition and produce relatively little smoke under fire exposure.

Resistance to insects and mould – The alkaline nature of lime-based binders provides protection against insects, mould and biological deterioration under suitable conditions.

Easy application and repair – Hempcrete can be used for infilling and renovation, and its lightweight nature makes construction and repair comparatively convenient.

4.2 DISADVANTAGES OF HEMPCRETE

Low compressive strength – Hempcrete has much lower compressive strength than conventional concrete and is therefore generally used as a non-load-bearing material.

High water absorption – Hempcrete can absorb considerable amounts of water. Prolonged exposure to moisture without adequate ventilation may create durability problems such as biological decay and freeze-thaw damage.

Long drying time – Hempcrete contains moisture during construction and requires adequate drying before finishes can be applied. High humidity can further delay drying.

Shrinkage – Shrinkage can occur during drying, and the binder type has a significant influ-

ence on the amount of shrinkage.

Requires skilled labour – Contractors and workers need knowledge of hempcrete mixing, placement, compaction and drying requirements. Lack of trained labour can restrict its wider use.

Limited availability – Hempcrete is not as widely available as conventional construction materials, and regulations related to hemp cultivation can affect its availability in some regions.

Lack of standardization – Standardized guidelines for hempcrete mix designs, applications and performance are still needed to ensure consistent quality.

CHAPTER 5
RESULTS AND DISCUSSION

5.1 COMPATIBILITY WITH TIMBER

Hempcrete is frequently associated with timber-frame construction because both materials can be incorporated into vapour-permeable wall assemblies. The review notes that hempcrete can help provide a protective environment for timber. However, exposure conditions and external cladding systems still require appropriate detailing.

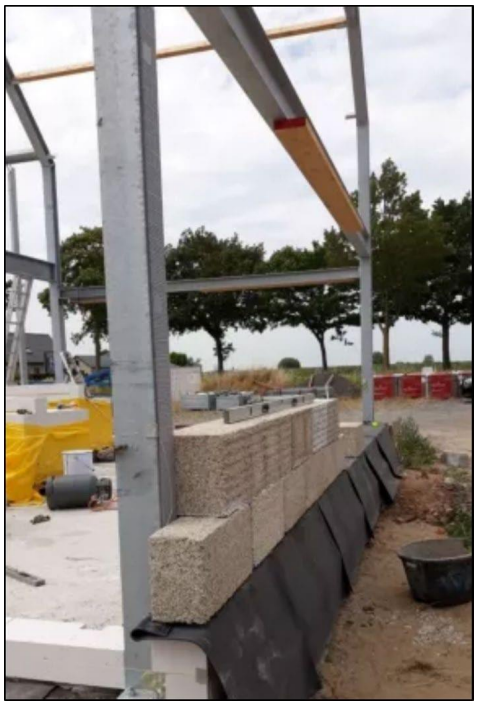


Fig 5.1 Hempcrete blocks utilised in a timber post-and-beam structure

5.2 COMPATIBILITY WITH MASONRY AND STEEL

Hempcrete can be cast against masonry walls as an insulating material. Existing gypsum plasters or cement renders may need to be removed where they are incompatible with a vapour-permeable lime-hemp assembly. Steel requires additional care because the alkaline environment associated with lime binders can create corrosion concerns. The review recommends separating steel frames from the hempcrete where practical, or protecting steel components with appropriate anti-corrosion measures.





(a) (b)

Fig 5.2 Hempcrete cast against masonry (a) and used with a metal structural system (b)

5.3 REAL-WORLD APPLICATION: HEMP HOTEL

The Hemp Hotel in Cape Town, South Africa, was completed in June 2023 and opened to the public in August 2023. The 12-storey building incorporates hempcrete as an infill material while concrete and cement provide the structural framework. The project is significant because it demonstrates the use of hemp-based material within a multi-storey urban building rather than limiting the technology to small low-rise structures.

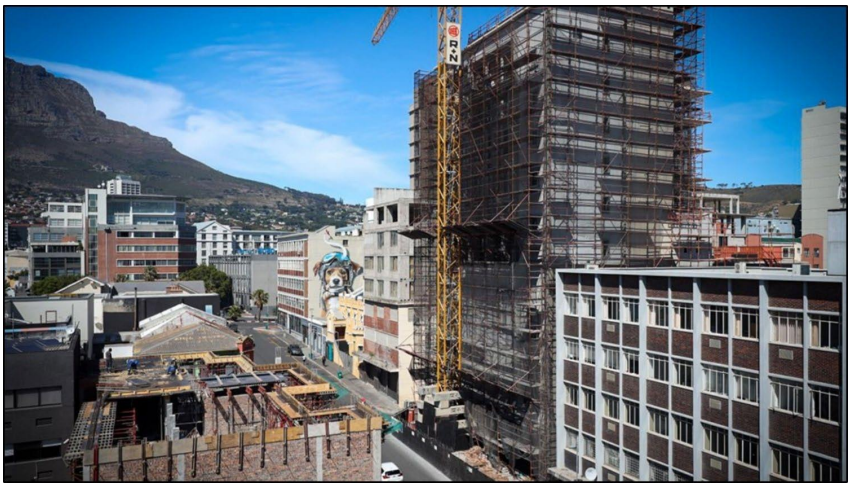


Fig 5.3 Hemp Hotel under construction, Cape Town

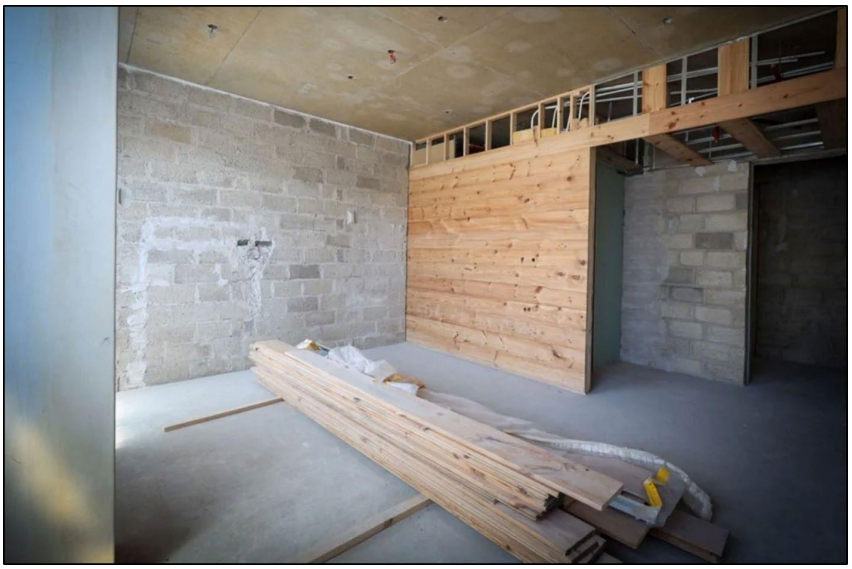


Fig 5.4 Interior construction of the Hemp Hotel

5.4 REAL-WORLD APPLICATION: ADNAMS BREWERS DISTRIBUTION CENTRE

The Adnams Brewers Distribution Centre in Suffolk, United Kingdom, represents a larger commercial application. The project used locally sourced hemp shives and lime for approximately 1000 m^3 of in-situ cast hempcrete together with roughly 100,000 hempcrete blocks. The project demonstrates that hempcrete can be deployed at a scale considerably beyond laboratory specimens.



(a)



(b)

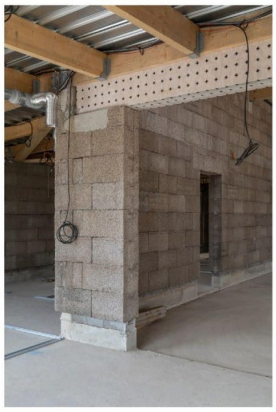
Fig 5.5 Adnams Brewers Distribution Centre: exterior (a) and in-situ casting (b)

5.5 PIERRE CHEVET SPORTS HALL

The Pierre Chevet Sports Hall in France was constructed in 2021 using hempcrete blocks and wood. The project is described in the principal review as France's first exclusive commercial hempcrete structure. Its significance lies in demonstrating how hempcrete blocks can form a major component of a commercial building system.



(a) (b)



(c)

Fig 5.6 Pierre Chevet Sports Hall, France

5.6 HYGROTHERMAL PERFORMANCE IN DIFFERENT CLIMATES

The review examines studies in France, Italy and Canada. In a hempcrete building in southwest France, reported indoor temperatures remained comparatively stable despite seasonal outdoor variations. A controlled climatic-chamber investigation in southern Italy also showed the ability of hempcrete walls to moderate internal humidity. A study comparing hempcrete and conventional wood-frame buildings in Vancouver and Toronto found that hempcrete reduced annual overheating hours in both climates.

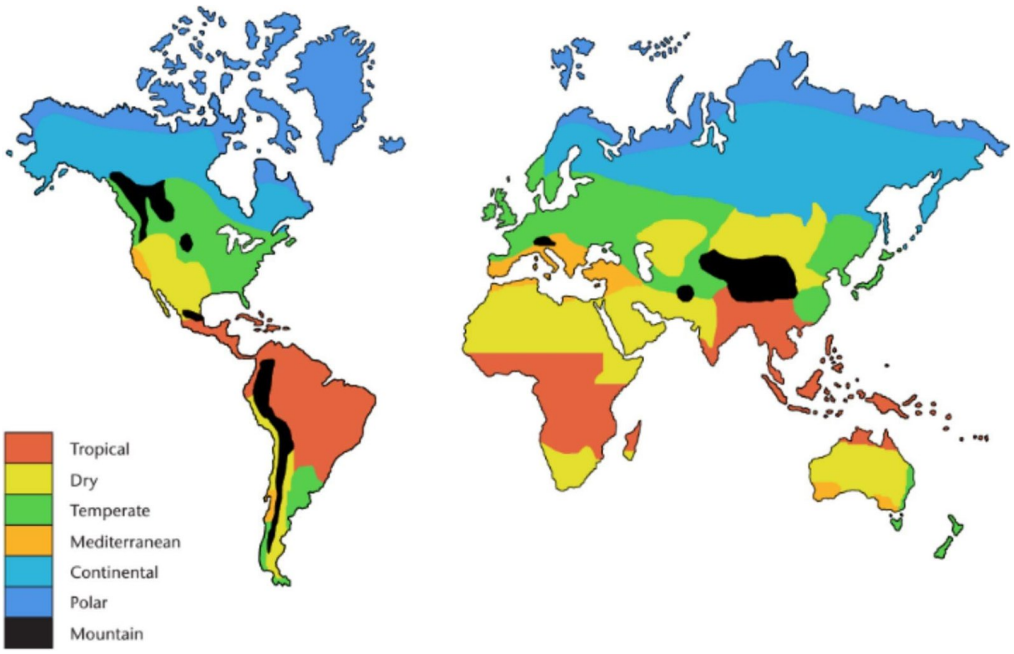


Fig 5.7 Climate zones represented in the reviewed performance studies

The review reports that hempcrete reduced overheating hours by 21.7% in Vancouver and 26.9% in Toronto. These findings indicate that hempcrete can moderate indoor temperature peaks under the investigated conditions, potentially improving occupant comfort and reducing reliance on air-conditioning systems.

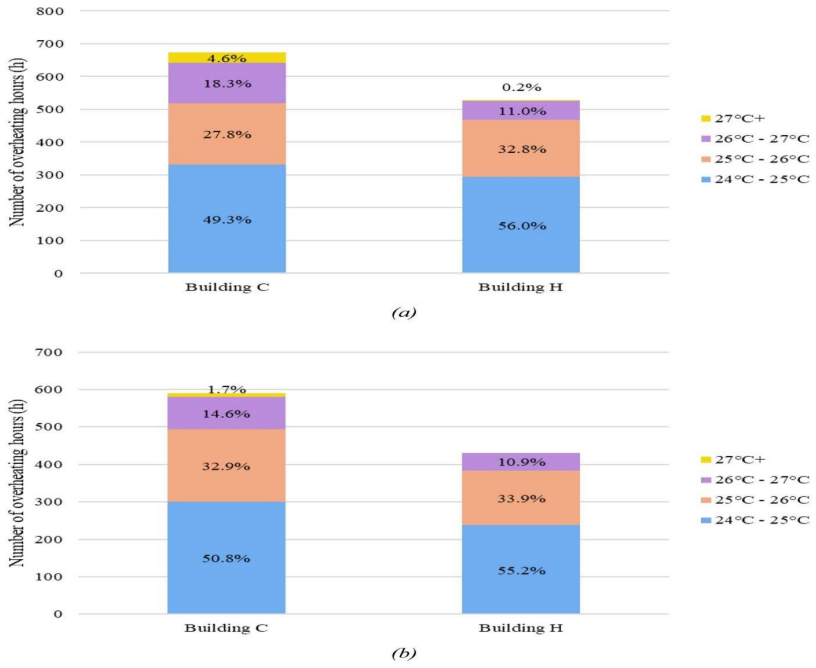


Fig 5.8 Annual overheating hours reported for the compared buildings

5.7 ENVIRONMENTAL PERFORMANCE

The environmental performance of hempcrete is closely connected to carbon storage. Hemp absorbs carbon dioxide during growth, and carbon remains embodied in the plant material after it becomes part of a building component. The lime binder can also absorb carbon dioxide during carbonation. Several life-cycle studies reviewed by Steyn et al. identify hempcrete as having potential to act as a carbon sink, but the magnitude of the benefit depends on the system boundary and assumptions.

5.8 LIFE-CYCLE ASSESSMENT

Life-cycle assessment must include cultivation, agricultural operations, processing, binder production, transport, construction, service life and end-of-life. The review notes that raw-material production, transportation distance and binder composition can strongly influence the environmental result. It is therefore inappropriate to describe hempcrete as automatically carbon-negative without defining the life-cycle boundary.

5.9 ECONOMIC CONSIDERATIONS

Cost depends on local hemp availability, processing infrastructure, binder supply, labour, equipment and construction familiarity. Some projects may have higher initial construction costs because specialist training or drying periods are required. However, the review also describes cases where improved indoor environmental control and lower operational energy can provide long-term economic benefits.

5.10 SCIENCE MUSEUM GROUP STORAGE FACILITY

The Science Museum Group storage facility in Wiltshire, United Kingdom, is another useful demonstration of hempcrete at building scale. The project incorporated precast hempcrete panels within a three-storey storage facility containing approximately 930 m^2 of storage space. The facility was designed to provide a stable and controlled indoor environment for archival collections, making hygrothermal performance particularly important.

The principal review reports that the facility required a longer drying period than initially anticipated. However, monitoring indicated that the hygrothermal behaviour of the hempcrete helped maintain suitable conditions for the stored collections during the stabilization phase. The review also cites a construction cost approximately 10% greater than conventional mu-

seum storage facilities, while operational costs were reported to be reduced by more than 50% compared with similarly sized climate-controlled spaces.

5.11 INTERPRETATION OF REAL-WORLD APPLICATIONS

The case studies demonstrate that hempcrete is not limited to experimental wall specimens. Its applications range from timber-frame houses and masonry retrofits to commercial distribution centres, sports halls, museum storage and multi-storey buildings. The common feature is that the structural and envelope roles are clearly coordinated. Hempcrete provides insulation, moisture regulation and other envelope functions while a suitable frame or supporting system carries the primary structural loads.

These projects also show why project-specific design is essential. Local climate, supply chains, drying conditions, labour skills and building regulations affect the feasibility of hempcrete. A successful demonstration in one country cannot simply be copied without considering the environmental and construction conditions of another region. For civil engineers, the correct approach is therefore to evaluate hempcrete as part of an integrated building system.

CHAPTER 6
CONCLUSION

6.1 CONCLUSION

- Hempcrete is a promising and sustainable building material that combines low density, thermal insulation, acoustic performance and moisture buffering.
- It offers biogenic carbon storage because hemp absorbs carbon dioxide while growing.
- Its performance depends on hemp-shive characteristics, binder chemistry, mix proportion, density and curing conditions.
- Its mechanical strength is low, so it is suited to non-load-bearing infill and insulation within a structural frame.
- Real-world projects show that it can be integrated with timber, masonry, steel and concrete systems.
- Moisture management, fire protection and quality control are essential.
- Overall, hempcrete is not a replacement for concrete but an effective low-carbon envelope material.

6.2 FUTURE SCOPE

Future research on hempcrete should focus on standardized mix designs and test methods, and on improving strength and ductility through optimized binders, additives and reinforcement. Long-term durability studies under realistic humidity cycles and whole-assembly fire studies are needed, together with life-cycle assessments using consistent system boundaries and economic studies of local supply chains and prefabricated components. Research on reuse and recycling, along with standardized design methods, specifications and regulations, will support the wider adoption of hempcrete as a reliable engineering material.

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